

Problem Set 3

Applied Stats/Quant Methods 1

Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in R using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
1 model_bivar_1 <- lm(voteshare ~ difflog, data = inc.sub)
2 summary(model_bivar_1)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.26832	-0.05345	-0.00377	0.04780	0.32749

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
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```

(Intercept) 0.579031    0.002251  257.19    <2e-16 ***
difflog      0.041666    0.000968   43.04    <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07867 on 3191 degrees of freedom
Multiple R-squared:  0.3673, Adjusted R-squared:  0.3671
F-statistic: 1853 on 1 and 3191 DF,  p-value: < 2.2e-16

```

A bivariate regression analysis was run by using the `lm` function in R. A bivariate regression was run because there was an outcome variable, `voteshare` and one explanatory variable, `difflog`

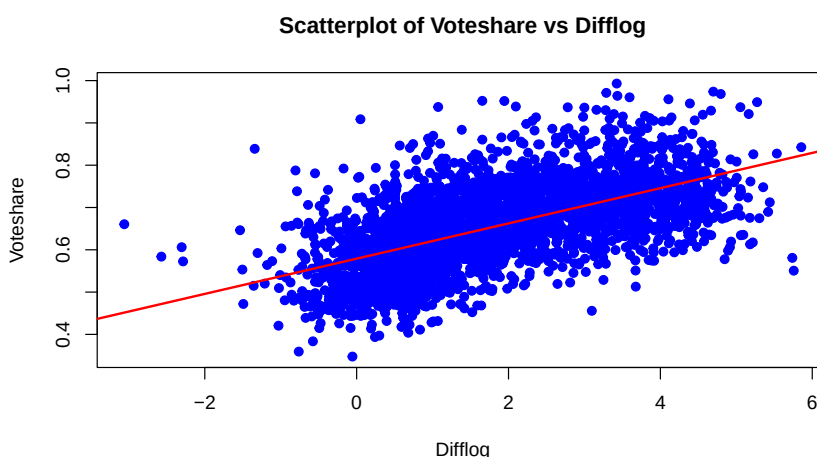
2. Make a scatterplot of the two variables and add the regression line.

```

1 plot(inc.sub$difflog, inc.sub$voteshare,
2       xlab = "Difflog", ylab = "Voteshare",
3       main = "Scatterplot of Voteshare vs Difflog",
4       pch = 19, col = "blue")
5 abline(lm(voteshare ~ difflog, data = inc.sub), col = "red", lwd = 2)

```

I created a scatter plot of the two variables that I would like to observe and also created a regression line through the scatter plot.



3. Save the residuals of the model in a separate object.

```

1 residuals_model_1 <- residuals(model_bivar_1)
2 head(residuals_model_1)

```

4. Write the prediction equation.

$$\widehat{voteshare}_i = \beta_0 + \beta_1 \text{difflog}_i$$

$$\text{presvote} = 0.579 + 0.0417 \times \text{difflog}$$

I used Latex code to do this.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is **presvote** and the explanatory variable is **difflog**.

```
1 model_bivar_2 <- lm(presvote ~ difflog, data = inc.sub)
2 summary(model_bivar_2)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.32196	-0.07407	-0.00102	0.07151	0.42743

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.507583	0.003161	160.60 <2e-16 ***
difflog	0.023837	0.001359	17.54 <2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1104 on 3191 degrees of freedom

Multiple R-squared: 0.08795, Adjusted R-squared: 0.08767

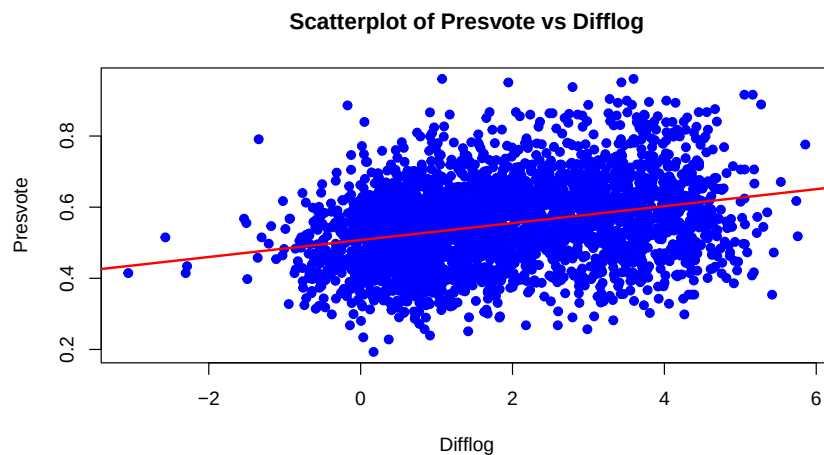
F-statistic: 307.7 on 1 and 3191 DF, p-value: < 2.2e-16

A bivariate regression analysis was run by using the lm function in R. A bivariate regression was run because there was an outcome variable, presvote and one explanatory variable, difflog

2. Make a scatterplot of the two variables and add the regression line.

```
1 plot(inc.sub$difflog, inc.sub$voteshare,
2       xlab = "Difflog", ylab = "Presvote",
3       main = "Scatterplot of Presvote vs Difflog",
4       pch = 19, col = "blue")
5 abline(lm(presvote ~ difflog, data = inc.sub), col = "red", lwd = 2)
```

I created a scatter plot of the two variables that I would like to observe and also created a regression line through the scatter plot.



3. Save the residuals of the model in a separate object.

```
1 residuals_model_2 <- residuals(model_bivar_2)
2 head(residuals_model_2)
```

4. Write the prediction equation.

$$\widehat{presvote}_i = \beta_0 + \beta_1 difflog_i$$

$$\text{presvote} = 0.507583 + 0.023837 \times \text{difflog}$$

I used Latex code to do this.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is **voteshare** and the explanatory variable is **presvote**.

```
1 model_bivar_3 <- lm(voteshare ~ presvote, data=inc.sub)
2 summary(model_bivar_3)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.27330	-0.05888	0.00394	0.06148	0.41365

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.441330	0.007599	58.08 <2e-16 ***
presvote	0.388018	0.013493	28.76 <2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08815 on 3191 degrees of freedom

Multiple R-squared: 0.2058, Adjusted R-squared: 0.2056

F-statistic: 827 on 1 and 3191 DF, p-value: < 2.2e-16

A bivariate regression analysis was run by using the lm function in R. A bivariate regression was run because there was an outcome variable, voteshare and one explanatory variable, presvote

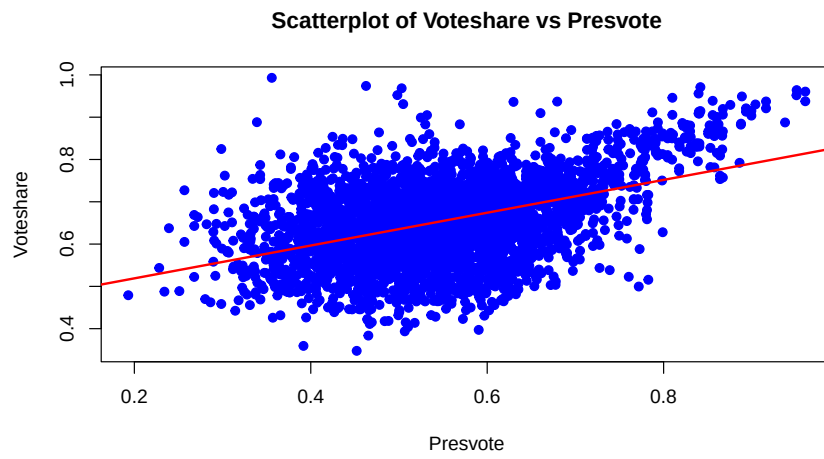
2. Make a scatterplot of the two variables and add the regression line.

```

1 plot(inc.sub$presvote, inc.sub$voteshare,
2       xlab = "Presvote",
3       ylab = "Voteshare",
4       main = "Scatterplot of Voteshare vs Presvote",
5       pch=19, col="blue")
6 model <- lm(voteshare ~ presvote, data= inc.sub)
7 abline(model, col="red", lwd=2)

```

I created a scatter plot of the two variables that I would like to observe and also created a regression line through the scatter plot.



3. Write the prediction equation.

$$\widehat{voteshare}_i = \beta_0 + \beta_1 presvote_i$$

$$voteshare = 0.441330 + 0.388018 \times presvote$$

I used Latex code to do this.

Question 4

The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
1 residual_model <- lm(residuals_model_1 ~ residuals_model_2)
2 summary(residual_model)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.25928	-0.04737	-0.00121	0.04618	0.33126

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.934e-18	1.299e-03	0.00
residuals_model_2	2.569e-01	1.176e-02	21.84

<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07338 on 3191 degrees of freedom

Multiple R-squared: 0.13, Adjusted R-squared: 0.1298

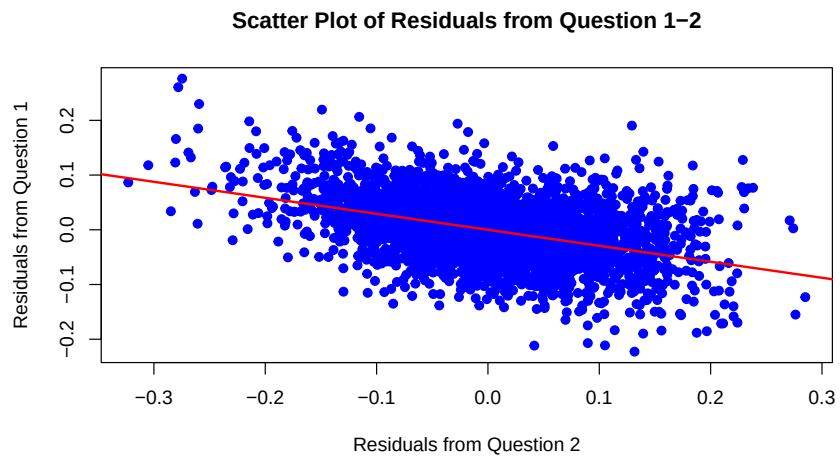
F-statistic: 477 on 1 and 3191 DF, p-value: < 2.2e-16

>

A bivariate regression analysis was run by using the lm function in R. A bivariate regression was run because there was an outcome variable, residuals from question 1 and one explanatory variable, the residuals from Question 2.

2. Make a scatterplot of the two residuals and add the regression line. *I created a scatter plot of the two variables that I would like to observe and also created a regression line through the scatter plot.*

```
1 plot(residuals_model_2, residuals_model_1,  
2       xlab = "Residuals from Question 2",  
3       ylab = "Residuals from Question 1",  
4       main = "Scatter Plot of Residuals from Question 1-2",  
5       pch = 19,  
6       col = "blue")  
7 abline(residual_model, col = "red", lwd = 2)
```



3. Write the prediction equation.

$$\text{residuals1} = \beta_0 + \beta_1 \cdot \text{residuals2}$$

$$\text{residuals_model_1} = -5.934 \times 10^{-18} + 0.2569 \times \text{residuals_model_2}$$

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
1 model_multi_4 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
2 summary(model_multi_4)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.25928	-0.04737	-0.00121	0.04618	0.33126

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.4486442	0.0063297	70.88 <2e-16 ***
difflog	0.0355431	0.0009455	37.59 <2e-16 ***
presvote	0.2568770	0.0117637	21.84 <2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07339 on 3190 degrees of freedom

Multiple R-squared: 0.4496, Adjusted R-squared: 0.4493

F-statistic: 1303 on 2 and 3190 DF, p-value: < 2.2e-16

This is a multivariate regression where multiple explanatory variables (difflog and presvote) are being compared against one outcome variable, vote share. I changed the regression analysis in R to reflect this.

2. Write the prediction equation.

For a multivariate analysis you include multiple variables in the equation.

$$\widehat{\text{votes}}_{\text{share}} = \beta_0 + \beta_1 \cdot \text{difflog} + \beta_2 \cdot \text{presvote} \quad (1)$$

$$\text{votes}_{\text{share}} = 0.4486442 + 0.0355431 \times \text{difflog} + 0.2568770 \times \text{presvote}$$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

The coefficient for the residual 2 is identical to the presvote coefficient. This value is 0.2568. This is because of the Frisch-Waugh-Lovell Theorem that states that the coefficient for a variable in a multivariate regression will be the same as a coefficient you would get from a two step process. In this problem set, I did a two step process where I regressed the dependent variable on the independent variable and then took the residual of that